

Question No:3

DS	Session	Searching	Question Information	Level 1	Challenge 3
Problem Description: Darsh recently found an new rectangular circuit board that he would like to recycle. That has R rows and C columns of squares. Each square of the circuit board has a very thinness, measured in millimetres. The square in the r-th row and c-th column has thinness $M_{r,c}$. A circuit board is beautiful if in each row, the difference between the thinnest square and the least thin square is no greater than L. Since the original circuit board might not be beautiful, Darsh would like to find a beautiful subcircuit board. This board can be obtained by choosing an axis-aligned subrectangle from the original board and taking the squares in that subrectangle. Darsh would like your help in finding the number of squares in the largest beautiful subrectangle of his original board. Constraints: $1 \leq T \leq 50$. $1 \leq R \leq 205$. $1 \leq C \leq 205$. $0 \leq M_i, j \leq 10^3$ for all i, j. $0 \leq L \leq 10^3$. Input Format: The first line of the input gives the number of test cases, T. T test cases follow. Each test case begins with one line containing three integers R, C and L, the number of rows, the number of columns, and the maximum difference in thinness allowed in each row. Then, there are R more lines containing C integers each. The c-th integer on the r-th line is M_r, c, the thinness of the square in the r-th row and c-th column. Output Format: Print the output in a separate lines contains the maximum number of squares in a beautiful subrectangle.					

▼ Logical Test Cases

Test Case 1

INPUT (STDIN)

3
1 4 0
3 1 3 3
2 3 0
4 4 5
7 6 6
4 5 0
2 2 4 4 20
8 3 3 3 12
6 6 3 3 3
1 6 8 6 4

EXPECTED OUTPUT

2
2
6

Test Case 2

INPUT (STDIN)

3
2 4 0
2 3 1 2
3 1 3 3
2 3 0
41 4 5
7 61 6
4 5 0
12 12 14 4 20
81 13 13 3 12
61 16 31 3 3
11 16 81 6 4

EXPECTED OUTPUT

2
2
4

▼ Mandatory Test Cases

Test Case 1

KEYWORD

int A[309][309];

Test Case 2

KEYWORD

bool ok[309][309][309];

▼ Complexity Test Cases

Test Case 1

Test Case 2

Test Case 3

Code:

```
1  #include <stdio.h>
2  #define MAXR 55
3  #define MAXC 210
4
5  long long a[MAXR][MAXC];
6
7  int main()
8  {
9      int T;
10     scanf("%d", &T);
11
12     while (T--)
13     {
14         int R, C;
15         long long L;
16
17         scanf("%d %d %lld", &R, &C, &L);
18
19         for (int i = 0; i < R; i++)
20         {
21             for (int j = 0; j < C; j++)
22             {
23                 scanf("%lld", &a[i][j]);
24             }
25         }
26
27         long long answer = 0;
28     }
```

```
29      /*
30         Fix the left column.
31         Extend the right column one by one.
32         For every row, maintain minimum and maximum
33         thickness in the current column range.
34     */
35     for (int left = 0; left < C; left++)
36     {
37         long long minVal[MAXR];
38         long long maxVal[MAXR];
39
40         for (int i = 0; i < R; i++)
41         {
42             minVal[i] = a[i][left];
43             maxVal[i] = a[i][left];
44         }
45
46         for (int right = left; right < C; right++)
47         {
48             if (right > left)
49             {
50                 for (int i = 0; i < R; i++)
51                 {
52                     if (a[i][right] < minVal[i])
53                         minVal[i] = a[i][right];
54                 }
```

```

55         maxVal[i] = a[i][right];
56     }
57 }
58 int consecutiveRows = 0;
59 int width = right - left + 1;
60
61 for (int i = 0; i < R; i++)
62 {
63     if (maxVal[i] - minVal[i] <= L)
64     {
65         consecutiveRows++;
66
67         long long area =
68             (long long)consecutiveRows * width;
69
70         if (area > answer)
71             answer = area;
72     }
73     else
74     {
75         consecutiveRows = 0;
76     }
77 }
78 printf("%lld\n", answer);
79 return 0;

```

Output:

Test Case-1:

```

3
1 4 0
3 1 3 3
2
2 3 0
4 4 5
7 6 6
2
4 5 0
2 2 4 4 20
8 3 3 3 12
6 6 3 3 3
1 6 8 6 4
6

```

Test Case-2:

```
3
2 4 0
2 3 1 2
3 1 3 3
2
2 3 0
41 4 5
7 61 6
2
4 5 0
12 12 14 4 20
81 13 13 3 12
61 16 31 3 3
11 16 81 6 4
4
PS C:\Users\kesha\OneDrive\Documents\3sem\C>
```

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